

VIEWPOINT

Informatics, Data Science, and Artificial Intelligence

Lisha Zhu, PhD

School of Biomedical Informatics, University of Texas Health Science Center, Houston.

W. Jim Zheng, PhD

School of Biomedical Informatics, University of Texas Health Science Center, Houston.

Biomedical research includes studies from the molecular level through population health and generates a wide range of data. To attain deeper insights into disease mechanisms, appropriate informatics approaches are needed to analyze these data at each level from small molecules to individuals and the entire population (Figure 1). For example, cheminformatics approaches are needed to analyze drug structure data and their therapeutic effects. Bioinformatics approaches are needed to analyze cellular and subcellular data and elucidate gene regulation. Imaging informatics are needed to gain diagnostic and prognostic insights into disease characteristics. The components of biomedical research, cross-scale data, and informatics constitute data science.

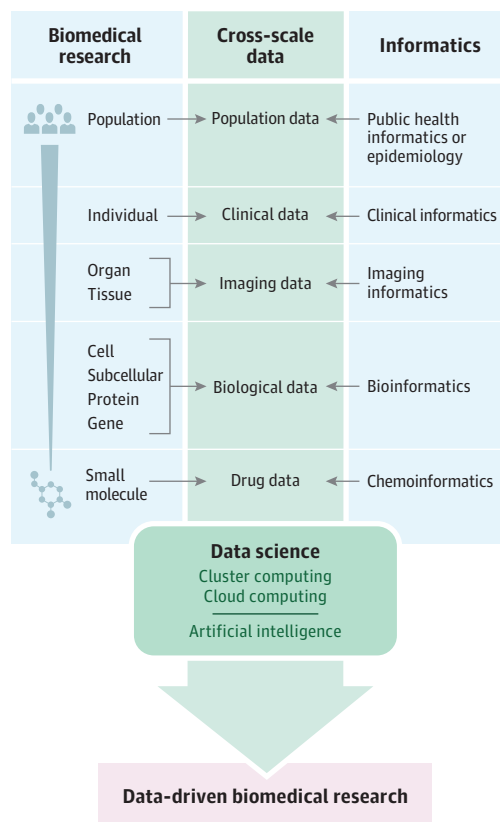
With the availability of electronic health records, clinical informatics approaches have been designed to find better cures, investigate disease comorbidities, and explore drug repurposing. Public health informatics or epidemiology is used to analyze diseases at the population level to help determine how diseases are transmitted and to detect patterns of disease in the population at large.¹

Even though each of these approaches has enabled remarkable advances in medicine, they are now challenged by recent advances in technology that have created enormous amounts of diverse data. For example, high-throughput screening and next-generation sequencing assays rapidly generate high volumes of data about drug effects, regulation of gene expression, and disease-relevant variants.

Furthermore, the wide adoption of electronic health records and their increasing use in research makes it possible to analyze the health data from millions of patients, ask new scientific questions, and better detect patterns to inform the development of new drugs or better care. Advances in computing technology have made possible the integration of this vast amount of cross-scale data at all levels of biomedical research, creating an even bigger challenge because of both the greater quantity of data and its increased complexity.

The discipline of data science has emerged in response to these new and evolving challenges. This new discipline seeks to contextualize and establish best practices for harnessing the recent and continuing advances in computing infrastructure, data processing, and analysis. One significant component of data science is use of complex computing infrastructures such as cluster computing (eg, Apache Hadoop and Spark) and cloud computing. These infrastructures are necessary to access, analyze, and rapidly manage cross-scale big data in biomedical research and enhance the use of artificial intelligence.

Figure 1. Biomedical Research and Informatics Approaches in Artificial Intelligence



However, to use big data effectively, scientists need to apply sophisticated algorithms to extract useful knowledge from so much heterogeneous information at all levels of biomedical research. This is where the latest advancements in artificial intelligence will be helpful.² Scientists have developed new artificial intelligence algorithms, such as deep neural networks, to analyze the high volume and complexity of big data.

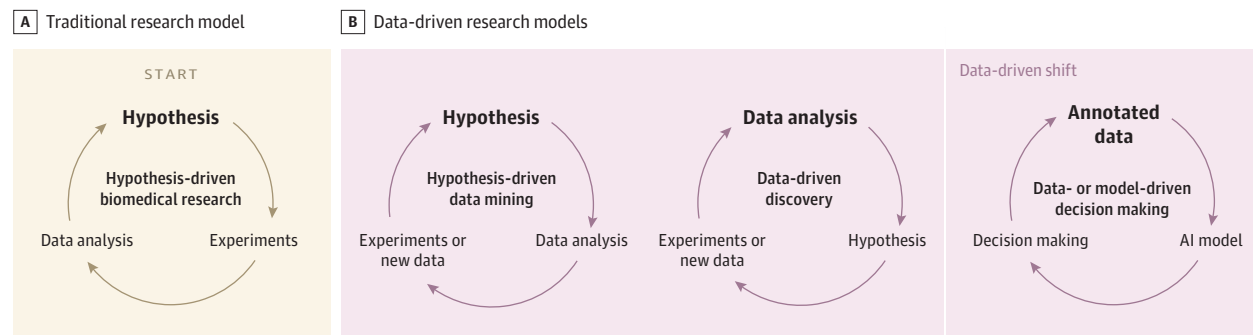
Applying advanced artificial intelligence algorithms in each informatics domain has shown promise to advance biomedical research. Two of many such examples include a deep learning algorithm to classify skin lesions, which performed as well as examinations by board-certified dermatologists,³ and new algorithms to help detect diabetic retinopathy.⁴

With data science and artificial intelligence, cross-scale biomedical research becomes feasible, but within a different model. Traditionally, most biomedical researchers use their knowledge to generate a hypothesis, design experiments or clinical studies to test the hypothesis, and use the above-mentioned informatics

Corresponding

Author: W. Jim Zheng, PhD, School of Biomedical Informatics, University of Texas Health Science Center, 7000 Fannin St, Ste 600, Houston, TX 77030 (wenjin.j.zheng@uth.tmc.edu).

Figure 2. Traditional Research Model and Data-Driven Models in Artificial Intelligence



approaches to perform a data analysis to test whether the study results support the hypothesis (Figure 2). With huge amounts of data becoming available, and advanced analysis tools such as artificial intelligence, biomedical research is driven more and more by data.

A research project can still be hypothesis driven, but instead of designing experiments, researchers can mine existing data to find evidence to support the hypothesis, and then design experiments or use additional data to further validate the hypothesis. Researchers can also start a project by mining existing data without an explicit hypothesis, and generate a testable hypothesis from the data mining results. Researchers can then conduct experiments or derive additional data to validate the hypothesis.

However, the most significant advancement is the approach driven by artificial intelligence models, which is different from traditional hypothesis-based research (Figure 2B). In this approach, large amounts of data are processed and annotated and then used to train an artificial intelligence model. The model is then used to evaluate new data and help make decisions.

Some of the most advanced artificial intelligence approaches, such as deep neural networks, do not require a thorough understanding of the underlying mechanism of such a model to achieve stunningly accurate results. This approach could be a valuable tool to interrogate cross-scale data and support clinical decisions even when many interconnections and relationships among these data are still unknown or missing.⁵

The availability of large amounts of interconnected data across all levels of biomedical research could enable thorough interrogation of relationships among those data, and may ultimately lead to breakthroughs that can improve health care. Although cross-scale analyses are not new in data analytic approaches, using well-annotated data to train advanced artificial intelligence models without much human intervention to provide valuable advice on therapeutic strategies and treatment outcomes for individual patients could bring significant breakthroughs—thereby potentially achieving the real promise of precision medicine approaches across multiple disciplines.

ARTICLE INFORMATION

Conflict of Interest Disclosures: Both authors have completed and submitted the ICMJE Form for Disclosure of Potential Conflicts of Interest. Dr Zheng reported receiving grants from the Cancer Prevention & Research Institute of Texas and the National Institutes of Health. No other disclosures were reported.

REFERENCES

1. Shortliffe EH, Cimino JJ. *Biomedical Informatics: Computer Applications in Health Care and Biomedicine*. 3rd ed. New York, NY: Springer; 2006.
2. Beam AL, Kohane IS. Big data and machine learning in health care. *JAMA*. 2018;319(13):1317-1318. doi:10.1001/jama.2017.18391
3. Esteva A, Kuprel B, Novoa RA, et al. Dermatologist-level classification of skin cancer with deep neural networks. *Nature*. 2017;542(7639):115-118. doi:10.1038/nature21056
4. Gulshan V, Peng L, Coram M, et al. Development and validation of a deep learning algorithm for detection of diabetic retinopathy in retinal fundus photographs. *JAMA*. 2016;316(22):2402-2410. doi:10.1001/jama.2016.17216
5. Weber GM, Mandl KD, Kohane IS. Finding the missing link for big biomedical data. *JAMA*. 2014;311(24):2479-2480.

Copyright of JAMA: Journal of the American Medical Association is the property of American Medical Association and its content may not be copied or emailed to multiple sites or posted to a listserv without the copyright holder's express written permission. However, users may print, download, or email articles for individual use.